

## 6.4 PRACTICAL AND IDEAL TRANSFORMERS

Transformers that are used in practice are called practical transformers. On the other hand, ideal transformers do not exist in reality but may be considered for theoretical studies only. Ideal transformers involve lots of approximation for the sake of simplicity and convenience in the analysis. A brief comparison of the two models is presented below in Table 6.1 for the sake of clarity.

Table 6.1 Comparison of practical and ideal transformers

| Practical transformer                                         | Ideal transformer                                       |
|---------------------------------------------------------------|---------------------------------------------------------|
| 1. Winding resistance is present.                             | 1. Winding resistance is neglected.                     |
| 2. Magnetic leakage is present.                               | 2. Magnetic leakage is neglected.                       |
| 3. Leakage reactance is present.                              | 3. Leakage reactance is neglected.                      |
| 4. Voltage drop in the windings is considered.                | 4. Voltage drop in the windings is neglected.           |
| 5. Terminal voltage differs from induced emf.                 | 5. Terminal voltage is same as induced emf.             |
| 6. Iron losses are present.                                   | 6. Iron losses may be neglected.                        |
| 7. Copper losses are present.                                 | 7. Copper losses are neglected.                         |
| 8. Efficiency is less than 100%.                              | 8. Efficiency is equal to 100%.                         |
| 9. Temperature rise takes place. Hence, cooling is necessary. | 9. No temperature rise. Hence, no cooling is necessary. |

## 6.5 TRANSFORMER LOSSES { constant loss + variable loss }

When connecting an alternating supply to the primary of a practical transformer, power loss takes place in the transformer irrespective of the secondary conditions (no-load or on-load). Two important losses in a practical transformer are iron loss and copper loss. Generally, iron loss in an electric machine is independent of load condition. Hence, it may be treated as constant loss for all operating conditions such as no-load, half-load, full-load, or any overload condition. However, copper loss being proportional to the square of the load current needs a very careful treatment while solving numerical problems. Assuming that iron loss irrespective of the load condition is represented as  $x$  and the copper loss at full-load condition (while delivering 100% of the rated output) is taken as  $y$ , at any fractional loading condition, the total loss may be considered as  $(x + k^2y)$ , where  $k$  is the ratio between the actual load prevailing at that time to the full load.

### 6.5.1 Iron Loss { CONSTANT LOSS }

If a transformer is operating in no-load condition, then the secondary does not draw any current, whereas the primary draws a small amount of no-load current to meet the magnetization requirements of the transformer. The loss that occurs in this condition is called *no-load loss*. As long as the supply voltage remains

constant, no-load loss remains fixed. Hence, no-load loss is also referred as constant or fixed loss. The no-load loss occurs mainly in the magnetic core of the transformer. Hence, it is also called *iron loss*. Iron loss is further divided into hysteresis loss and eddy current loss. The main reason behind these losses is the continuous magnetic reversal in the core as imposed by the alternating mmf. The expressions for hysteresis loss and eddy current loss are given in Eqns (6.15) and (6.16) respectively.

**Hysteresis loss** The hysteresis loss is given by

$$P_h = K_h f (B_m)^k \quad (6.15)$$

In this expression,

$P_h$  = Hysteresis loss per unit volume in Watts/m<sup>3</sup>

$K_h$  = Hysteresis coefficient (varies from material to material)

$f$  = Cyclic frequency of magnetization (hertz)

$B_m$  = Maximum flux density (Wb/m<sup>2</sup>)

$k$  = Steinmetz coefficient (varies between 1.5 to 2.5)

**Eddy current loss** The eddy current loss is given by

$$P_e = K_e f^2 B_m^2 \quad (6.16)$$

In this expression,

$P_e$  = Eddy current loss per unit volume in Watts/m<sup>3</sup>

$K_e$  = Eddy current loss coefficient

$f$  = Cyclic frequency of magnetization (hertz)

$B_m$  = Maximum flux density (Wb/m<sup>2</sup>)

Finally, it may be written that

Iron loss = Hysteresis loss + Eddy current loss

$$P_i = (P_h + P_e) W \quad (6.17)$$

These two components of iron loss contribute towards temperature rise of the core, and as a result, the performance of the transformer gets affected. There are some schemes available in practice to limit the iron loss in a practical transformer. These schemes come under the design aspects and are incorporated at the time of transformer manufacturing.

Hysteresis loss cannot be totally eliminated but can be reduced significantly by use of highly permeable magnetic material in the core (usually silicon grade-grain oriented steel). Similarly, eddy current loss cannot be totally eliminated but can be reduced significantly by laminating the magnetic material used in the core. Thinner laminations would result in lesser eddy current loss.

### 6.5.2 Copper Loss { Variable loss }

As the secondary of a practical transformer gets connected with a load device, a secondary current ( $I_2$ ) and an equivalent primary current ( $I_1$ ) flow in the respective windings. The resistance present in the windings offers opposition to the flow of current in the windings. However, to overcome this opposition, power loss takes

place in the copper windings in the form of  $I^2R$  loss. Hence, this loss is also called *ohmic loss* or *copper loss*. The total amount of copper loss in a transformer is represented in Eqn (6.18) as the algebraic sum of the two copper losses.

$$P_{Cu} = \text{Primary copper loss} + \text{Secondary copper loss} \\ = I_1^2 R_1 + I_2^2 R_2 \quad \text{W} \quad (6.18)$$

An inverted tree chart for highlighting the classification of transformer losses is shown in Fig. 6.6.

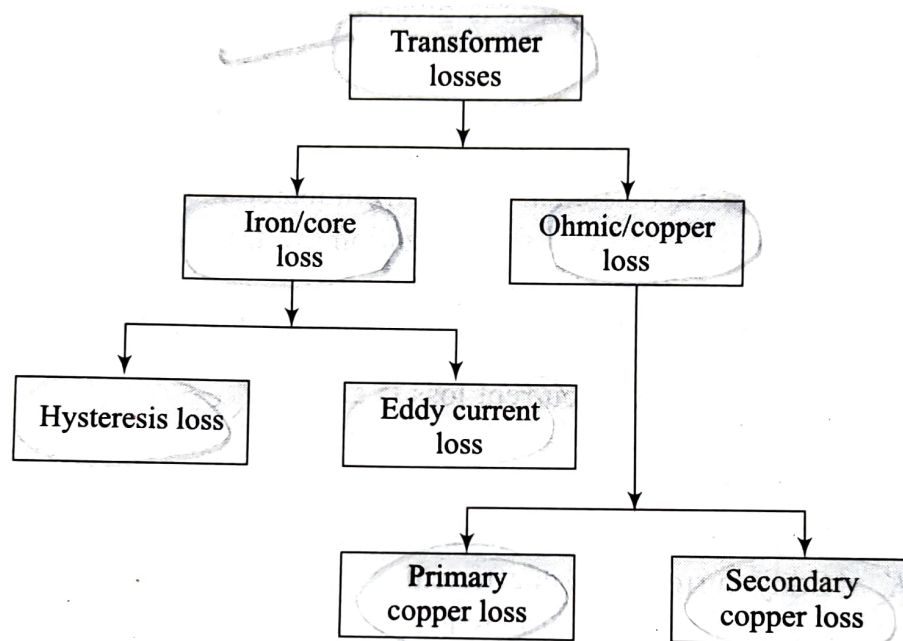


Fig. 6.6 Classification of transformer losses

## 6.6 TRANSFORMER PHASOR DIAGRAM

It is understood by now that the main electrical quantities required to be involved for a satisfactory transformer operation are (a) alternating supply voltage, (b) alternating current, and (c) alternating magnetic flux. These quantities can be represented in two forms—waveform representation and phasor representation—as has already been explained in Chapter 4. Representation of the alternating flux, alternating emf's, alternating voltages, and alternating currents corresponding to both sides of a transformer in phasor form is referred as *transformer phasor diagram*. There may be two types of phasor diagrams depending on the operation of transformers—no-load phasor diagram and on-load phasor diagram.

### 6.6.1 No-Load Phasor Diagram

In order to draw the no-load phasor diagram, we start with flux phasor as the reference phasor and plot it along the positive  $x$ -axis of the coordinate system as shown in Fig. 6.7(b). The corresponding circuit diagram is shown in Fig. 6.7(a).

As already indicated in Eqns (6.4), (6.5) and (6.7), a sinusoidal flux induces emf's in primary and secondary windings, which are also sinusoidal in nature